

In Class Exercise

March 30, 2026

An offshore wind farm consists of turbines and transfer stations that must be connected to power stations on shore through underwater cables. Each cable has:

- A fixed installation cost (based on distance)
- A variable cost depending on the amount of electrical flow it carries

The goal is to design a network that connects all turbines to the shore at minimum cost.

Sets and Indices

- $G(V, E)$: Graph representing the wind farm network
 - V : set of vertices (turbines, transfer stations, power stations)
 - E : set of potential cable connections

Parameters

- $s_i \in \mathbb{R}$: Power supplied at node $i \in V$
 - $s_i > 0$ for turbines (sources)
 - $s_i = 0$ for transfer stations
 - $s_i < 0$ for power stations (sinks)
- $u_{ij} \in \mathbb{R}^+$: Capacity of cable (i, j)
- $c_{ij} \in \mathbb{R}^+$: Cost per unit of flow on arc (i, j)
- $f_{ij} \in \mathbb{R}^+$: Fixed cost of installing cable (i, j)

Data

Let's consider an example with three turbines, one transfer station, and two power stations.

Node Supplies/Demands (MW)

Vertex i	s_i
0	4
1	3
2	2
3	0
4	-6
5	-3

Edge Data

Edge (i, j)	u_{ij}	c_{ij}	f_{ij}
(0,4)	4	1	1
(0,3)	2	1	1
(1,3)	3	1	1
(2,5)	2	1	1
(3,4)	2	1	1
(3,5)	1	1	1

In Class Exercise

March 30, 2026

Tasks

1. Network Representation:

- Draw the graph $G(V, E)$ using the data above.
- Label nodes with s_i and edges with (u_{ij}, c_{ij}, f_{ij}) .

2. MIP Formulation:

- Define decision variables for:
 - Flow on each edge
 - Whether a cable is installed
- Formulate an optimization model that minimizes total cost.
- Include:
 - Flow conservation constraints
 - Capacity constraints
 - Logical constraints linking flow and installation decisions